MediQore

Clinical Rules Table — Review Checklist for the Clinical Advisor

Table version 0.3.0 · status pending clinical review · prepared 22 October 2026

MediQore is a final-year project (BS Computer Science, COMSATS University Islamabad) by Muhammad Zain Abbas and Zain Ali: an offline Android app in Urdu and English for Pakistan's Lady Health Workers, with a web portal for supervisors. Every clinical threshold the app uses — danger signs, blood pressure and haemoglobin levels, obstetric history flags, typing-error ranges, EPI timings, MUAC and IMCI rules — is kept in one versioned table, never inside the program code.

The values below are WHO-referenced defaults chosen by the student team. No clinician has reviewed them yet. The prototype runs only on synthetic (made-up) data. The table may be used with real patients only after a Clinical Advisor has reviewed and signed it. The app is decision support only; it never replaces the LHW's referral or a clinician's judgement.

How to review (about 60–90 minutes)

1. For each numbered item, tick Agree or Change. For a change, write the value you recommend in the comments box of that section.
2. Answer each amber Decision needed box. These are values the project scope leaves to the Clinical Advisor.
3. Initial every page at the bottom, then complete and sign the sign-off page at the end.
4. Return the checklist to the team. They update the table, raise its version to 1.0.0, mark it signed with your name and date, and run the automated tests that check every rule.

How the rules are applied in the app

• A danger sign gives an Emergency (automatically Red). The final risk level is always the higher of the AI model's result and the rules' result, so a rule can raise the level but never lower it.
• Flags (obstetric history, swelling, anaemia) are shown to the LHW but do not change the risk colour unless you decide they should.
• Units are fixed: BP mmHg, pulse beats/min, temperature °C, blood sugar mmol/L, Hb g/dL, weight kg, MUAC mm.

A. Visit form: typing-error checks

These ranges only catch typing mistakes on the visit form; they do not decide risk.

#	Item	Proposed value	Source	Decision
1	**Systolic BP**	Usual range 60–250 mmHg (outside: the LHW must confirm) Possible range 20–300 mmHg (outside: cannot be saved)	Scope	☐ Agree ☐ Change
2	**Diastolic BP**	Usual range 30–150 mmHg (outside: the LHW must confirm) Possible range 10–200 mmHg (outside: cannot be saved)	Team proposal	☐ Agree ☐ Change
3	**Weight**	Usual range 30–150 kg (outside: the LHW must confirm) Possible range 10–250 kg (outside: cannot be saved)	Team proposal	☐ Agree ☐ Change
4	**Temperature**	Usual range 34–42 °C (outside: the LHW must confirm) Possible range 25–45 °C (outside: cannot be saved)	Team proposal	☐ Agree ☐ Change
5	**Pulse**	Usual range 40–180 beats/min (outside: the LHW must confirm) Possible range 20–250 beats/min (outside: cannot be saved)	Team proposal	☐ Agree ☐ Change
6	**Blood sugar**	Usual range 2–30 mmol/L (outside: the LHW must confirm) Possible range 0.5–50 mmol/L (outside: cannot be saved)	Team proposal	☐ Agree ☐ Change
7	**Systolic BP: normal range**	90–139 mmHg (outside: the value is highlighted in amber)	WHO PCPNC 2015	☐ Agree ☐ Change
8	**Diastolic BP: normal range**	60–89 mmHg (outside: the value is highlighted in amber)	WHO PCPNC 2015	☐ Agree ☐ Change
9	**Temperature: normal range**	36.0–37.9 °C (outside: the value is highlighted in amber)	WHO PCPNC 2015	☐ Agree ☐ Change
10	**Pulse: normal range**	60–100 beats/min (outside: the value is highlighted in amber)	Confirmed by the Clinical Advisor, 22 Oct 2026	☐ Agree ☐ Change
11	**Pulse counter**	Count beats for 30 s, multiply by 2	Scope	☐ Agree ☐ Change
12	**Blood sugar conversion**	mmol/L = mg/dL ÷ 18.0	Standard	☐ Agree ☐ Change

Changes or comments for this section:

Reviewer initials: _____

B. Pregnant woman: danger signs checked after every visit

Evaluated on the phone, offline, independently of the AI model. Any Emergency opens the emergency screen (SMS and call to the supervisor).

#	Item	Proposed value	Source	Decision
1	**Systolic BP 160 mmHg or above, or diastolic BP 110 mmHg or above.**	Emergency (Red)	WHO MCPC 2017	☐ Agree ☐ Change
2	**BP 140/90 mmHg or above together with severe headache, blurred vision or severe/upper abdominal pain.**	Emergency (Red)	WHO MCPC 2017	☐ Agree ☐ Change
3	**Vaginal bleeding.**	Emergency (Red)	WHO PCPNC 2015	☐ Agree ☐ Change
4	**Convulsions or fits.**	Emergency (Red)	WHO PCPNC 2015	☐ Agree ☐ Change
5	**Severe headache with blurred vision.**	Emergency (Red)	WHO PCPNC 2015	☐ Agree ☐ Change
6	**Severe or upper abdominal pain.**	Emergency (Red)	WHO PCPNC 2015	☐ Agree ☐ Change
7	**Fast or difficult breathing.**	Emergency (Red)	WHO PCPNC 2015	☐ Agree ☐ Change
8	**Fever and too weak to get out of bed.**	Emergency (Red)	WHO PCPNC 2015	☐ Agree ☐ Change
9	**Temperature 38 °C or above together with another danger sign.**	Emergency (Red)	WHO PCPNC 2015	☐ Agree ☐ Change
10	**Absent or markedly reduced fetal movement. The form's 'reduced' option means markedly reduced.**	Emergency (Red)	WHO PCPNC 2015	☐ Agree ☐ Change
11	**Severe anaemia signs (for example very pale palms or inner eyelids with breathlessness).**	Emergency (Red)	WHO PCPNC 2015	☐ Agree ☐ Change
12	**BP 140/90 mmHg or above on its own sets the level to at least Yellow, with a facility check.**	At least Yellow, facility check	WHO MCPC 2017	☐ Agree ☐ Change
13	**Swelling is recorded and shown as a flag; it does not change the risk level.**	Flag only, risk unchanged	WHO PCPNC 2015	☐ Agree ☐ Change
14	**AI model classes**	Low → Green, Mid → Yellow, High → Red	Project design	☐ Agree ☐ Change
15	**Fetal movement question**	First felt between 16 and 22 weeks; the visit form asks from 22 weeks, earlier it records no answer	Confirmed by the Clinical Advisor, 22 Oct 2026	☐ Agree ☐ Change
16	**Expected delivery date**	LMP + 280 days; from the month told: month m is about (m − 0.5) × 4.345 weeks; registration allowed from 1 to 44 weeks	Naegele's rule; team proposal	☐ Agree ☐ Change

Decision needed: Raised BP (140/90 or above) on its own gives Yellow with a facility check. Within how many hours should the woman be seen at a facility?
☐ 24 hours ☐ 48 hours ☐ 72 hours ☐ Other: ____
Answer: _____

Changes or comments for this section:

C. Anaemia (verified haemoglobin) and obstetric history flags

#	Item	Proposed value	Source	Decision
1	**Anaemia: Severe**	Hb below 7.0 g/dL	WHO Hb 2011	☐ Agree ☐ Change
2	**Anaemia: Moderate**	Hb 7.0 to below 10.0 g/dL	WHO Hb 2011	☐ Agree ☐ Change
3	**Anaemia: Mild**	Hb 10.0 to below 11.0 g/dL	WHO Hb 2011	☐ Agree ☐ Change
4	**Anaemia: No anaemia**	Hb 11.0 g/dL or above	WHO Hb 2011	☐ Agree ☐ Change
5	**Anaemia flag changes risk colour?**	No (flag only)	Scope	☐ Agree ☐ Change
6	**Obstetric flag: One or more previous C-sections.**	Shown on the profile; risk unchanged	WHO MCPC 2017	☐ Agree ☐ Change
7	**Obstetric flag: One or more previous stillbirths.**	Shown on the profile; risk unchanged	WHO ANC 2016	☐ Agree ☐ Change
8	**Obstetric flag: Five or more previous pregnancies.**	Shown on the profile; risk unchanged	WHO MCPC 2017	☐ Agree ☐ Change
9	**Obstetric flag: A known medical condition was recorded.**	Shown on the profile; risk unchanged	WHO ANC 2016	☐ Agree ☐ Change
10	**Obstetric flag: Age under 18 at registration.**	Shown on the profile; risk unchanged	WHO ANC 2016	☐ Agree ☐ Change
11	**Obstetric flag: Age 35 or above at registration.**	Shown on the profile; risk unchanged	WHO ANC 2016	☐ Agree ☐ Change

Decision needed: WHO 2011 sets mild anaemia below 11.0 g/dL throughout pregnancy; the 2024 WHO guideline uses 10.5 g/dL in the second trimester. Which should the app apply?
☐ WHO 2011 (11.0) ☐ WHO 2024 (10.5 in 2nd trimester)
Answer: _____

Decision needed: Only 'Previous C-section' is named in the scope. Should the other obstetric flags and cut-offs above stay (five or more pregnancies, under 18, 35 or above, previous stillbirth, known condition)? Should any of them raise the risk colour?
☐ Keep all, flags only ☐ Change (write below)
Answer: _____

Changes or comments for this section:

Reviewer initials: _____

D. Antenatal care schedule, tetanus toxoid and supplements

#	Item	Proposed value	Source	Decision
1	**Stillbirth definition**	Pregnancy ending without a live birth at 28 completed weeks or later; earlier is a miscarriage	WHO 2016	☐ Agree ☐ Change
2	**ANC contacts**	First contact up to 12 weeks; then at 20, 26, 30, 34, 36, 38, 40 weeks (8 contacts)	WHO ANC 2016	☐ Agree ☐ Change
3	**TT/Td dose 1**	At first contact	WHO TT 2017	☐ Agree ☐ Change
4	**TT/Td dose 2**	At least 4 weeks after the previous dose	WHO TT 2017	☐ Agree ☐ Change
5	**TT/Td dose 3**	At least 6 months after the previous dose	WHO TT 2017	☐ Agree ☐ Change
6	**TT/Td dose 4**	At least 12 months after the previous dose	WHO TT 2017	☐ Agree ☐ Change
7	**TT/Td dose 5**	At least 12 months after the previous dose	WHO TT 2017	☐ Agree ☐ Change
8	**Iron and folic acid**	Daily: iron 30–60 mg, folic acid 400 µg	WHO ANC 2016	☐ Agree ☐ Change

Changes or comments for this section:

E. Routine child immunisation (EPI), children under 2

Schedule version pk-epi-2022.1, effective 2022-01-01. A late child's next dose is due at the later of the due age and the previous dose plus the minimum interval.

#	Item	Proposed value	Source	Decision
1	**BCG-1**	Due at 0 days; earliest 0 days; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
2	**OPV-0**	Due at 0 days; earliest 0 days; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
3	**HepB-0**	Due at 0 days; earliest 0 days; latest ____ (to fill); only in areas where it is introduced	NIP Pakistan 2022	☐ Agree ☐ Change
4	**OPV-1**	Due at 6 weeks; earliest 6 weeks; at least 4 weeks after previous dose; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
5	**Penta-1**	Due at 6 weeks; earliest 6 weeks; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
6	**PCV-1**	Due at 6 weeks; earliest 6 weeks; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
7	**Rota-1**	Due at 6 weeks; earliest 6 weeks; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
8	**OPV-2**	Due at 10 weeks; earliest 10 weeks; at least 4 weeks after previous dose; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
9	**Penta-2**	Due at 10 weeks; earliest 10 weeks; at least 4 weeks after previous dose; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
10	**PCV-2**	Due at 10 weeks; earliest 10 weeks; at least 4 weeks after previous dose; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
11	**Rota-2**	Due at 10 weeks; earliest 10 weeks; at least 4 weeks after previous dose; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
12	**OPV-3**	Due at 14 weeks; earliest 14 weeks; at least 4 weeks after previous dose; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
13	**Penta-3**	Due at 14 weeks; earliest 14 weeks; at least 4 weeks after previous dose; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
14	**PCV-3**	Due at 14 weeks; earliest 14 weeks; at least 4 weeks after previous dose; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
15	**IPV-1**	Due at 14 weeks; earliest 14 weeks; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
16	**MR-1**	Due at 9 months; earliest 9 months; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
17	**TCV-1**	Due at 9 months; earliest 9 months; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
18	**IPV-2**	Due at 9 months; earliest 9 months; at least 4 months after previous dose; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change

#	Item	Proposed value	Source	Decision
19	**MR-2**	Due at 15 months; earliest 15 months; at least 4 weeks after previous dose; latest ____ (to fill)	NIP Pakistan 2022	☐ Agree ☐ Change
20	**Zero-dose child**	No recorded Penta-1 after its due age	NIP Pakistan 2022	☐ Agree ☐ Change

Decision needed: Please check every row against the current Federal Directorate of Immunization schedule and fill in the 'latest' (maximum) age where a dose stops being given, for example rotavirus and OPV-0.
Answer: _____

Changes or comments for this section:

F. Child nutrition (MUAC) and sick-child assessment (IMCI, 2 months to 5 years)

#	Item	Proposed value	Source	Decision
1	**SAM (age 6–59 months)**	MUAC below 115 mm	WHO/UNICEF 2009	☐ Agree ☐ Change
2	**MAM (age 6–59 months)**	MUAC 115 to below 125 mm	WHO/UNICEF 2009	☐ Agree ☐ Change
3	**Normal (age 6–59 months)**	MUAC 125 mm or above	WHO/UNICEF 2009	☐ Agree ☐ Change
4	**Bilateral pitting oedema**	SAM regardless of MUAC	WHO SAM 2013	☐ Agree ☐ Change
5	**Underweight**	Weight-for-age below -2 SD; severe below -3 SD	WHO Growth 2006	☐ Agree ☐ Change
6	**Stunting**	Height-for-age below -2 SD; severe below -3 SD (only when height is measured)	WHO Growth 2006	☐ Agree ☐ Change
7	**Fast breathing, 2–11 months**	50 breaths/min or more, counted for 60 s	WHO IMCI 2014	☐ Agree ☐ Change
8	**Fast breathing, 12–59 months**	40 breaths/min or more, counted for 60 s	WHO IMCI 2014	☐ Agree ☐ Change
9	**Severe pneumonia or very severe disease**	When any general danger sign → Pink: refer urgently	WHO IMCI 2014	☐ Agree ☐ Change
10	**Pneumonia**	When chest indrawing or fast breathing → Yellow: treat	WHO IMCI 2014	☐ Agree ☐ Change
11	**Cough or cold**	When no signs of pneumonia or very severe disease → Green: home care	WHO IMCI 2014	☐ Agree ☐ Change
12	**Diarrhoea**	3 or more loose stools per day	WHO IMCI 2014	☐ Agree ☐ Change
13	**Severe dehydration**	2 or more of: lethargic or unconscious, sunken eyes, not able to drink or drinking poorly, skin pinch very slow → Pink: refer urgently	WHO IMCI 2014	☐ Agree ☐ Change
14	**Some dehydration**	2 or more of: restless or irritable, sunken eyes, drinks eagerly thirsty, skin pinch slow → Yellow: treat	WHO IMCI 2014	☐ Agree ☐ Change
15	**No dehydration**	not enough signs for the classes above → Green: home care	WHO IMCI 2014	☐ Agree ☐ Change

Decision needed: IMCI general danger signs listed for the checklist: unable to drink or breastfeed, vomits everything, convulsions, lethargic or unconscious. Is this list complete for LHW use?
☐ Yes ☐ Add or remove (write below)
Answer: _____

Changes or comments for this section:

G. Sign-off

I have reviewed sections A to F of the MediQore Clinical Rules Table version 0.3.0. Subject to the changes and decisions I have written in this checklist, I approve these rules as decision support for Lady Health Workers. I understand that the app does not replace a clinician's judgement, and that the AI risk model has not yet been validated on Pakistani field data.

☐ Approved without changes ☐ Approved with the changes written in this checklist ☐ Not approved

Name	
Qualification and speciality	
PMDC registration number	
Hospital or institution	
Phone or email	
Date	
Signature	

For the team after signing

1. Apply every change and decision to clinical-rules/clinical-rules.json and remove the review notes.
2. Set status to 'signed', fill sign_off (name, qualification, date) and raise the version to 1.0.0.
3. Update test-cases.json, copy the table into the app, run all tests, and keep this signed checklist with the project records (not in the public repository if it shows personal contact details).